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Cloud Computing

BCA 5th Semester, Batch BCA2024
School of Engineering

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Cloud Storages

Three primary types of storages are:

1. Object Storage. Ex: Amazon S3 (Simple Storage Service)
2. Block Storage. Ex: Amazon EBS (Elastic Block Storage)
3. File Storage. Ex: Amazon EFS (Elastic File System)

Cloud Storages

Storage Types Across Major Service Providers

Storage Type	AWS	Azure	Google Cloud (GCP)
Block Storage	**Amazon EBS** (Elastic Block Store)	**Azure Disk Storage**	**Persistent Disk, Hyperdisk**
File Storage	**Amazon EFS** (Elastic File System), **FSx**	**Azure Files**, **Azure NetApp Files**	**Filestore**, **NetApp Volumes**
Object Storage	**Amazon S3** (Simple Storage Service)	**Azure Blob Storage**	**Google Cloud Storage**

Key Differences

- **Block Storage** → Best for **VM disks, databases, and transactional workloads**.
- **File Storage** → Best for **shared access, enterprise apps**.
- **Object Storage** → **Best for archives**, **scalable data lakes, backups, media, and analytics**.

Object Storage

What is object storage? Object storage, also called object-based storage, is a solution that organizes and manages data as individual units known as objects. Each object contains data, metadata, and a unique identifier, allowing it to be stored independently in a flat, scalable system, unlike the **hierarchical folder structure** used in traditional file storage. Object storage is ideal for storing enormous amounts of unstructured data. **Access through API**, which can be used by users or applications.

With object storage in the cloud, this architecture ensures that all data is organized within a single addressable storage pool, making it more accessible and scalable, especially for large-scale unstructured data.

- **Amazon S3** is a **simple storage service** offered by Amazon and it is useful for hosting website images and videos, data analytics, etc. S3 is an **object-level data storage** that distributes the data objects across several machines and allows the users to access the storage via the internet from any corner of the world.

Object Storage

Object Storage Functions

Object storage offers various extensible properties that can be configured. These may include:

- **Logging** – monitors (logs) operations in the storage
- **Lifecycle Management** - automates deletion or moving of objects (or object versions) to cheaper storage classes
- **S3 Versioning** - retains multiple versions of an object
- **S3 Object Lock** – prevents permanent deletion of objects

Block Storage

What is Block Storage? Block storage is a data storage solution where data is divided into smaller blocks and stored across multiple physical drives. Each block has a unique address, making block storage ideal for high-performance applications like databases.

Amazon EBS(elastic block storage) is a **block-level data storage** offered by Amazon. Block storage stores files in multiple volumes called blocks, which act as separate hard drives, and this storage is not accessible via the internet. Use cases include business continuity, transactional and NO SQL database, software testing, etc. These are primarily used for attaching to EC2 instances.

- SAN (Storage Area Network) Storage

File System Storage

EFS is a file storage system. File storage is the system you'll likely be most familiar with, as it's how files are stored in the hard drive on your computer. File storage is fast and accessible, but it doesn't offer the increased potential for complex queries that object storage does. Unlike EBS, EFS can be mounted by multiple EC2 instances, meaning many virtual machines may store files within an EFS instance.

NAS (Network Attached Storage) Storage

EFS's key benefits

- Adaptive throughput – EFS's performance can scale in-line with its storage, operating at a higher throughput for sudden, high-volume file dumps, reaching up to 500,000 IOPS or 10 GB per second
- Totally elastic – once you've spun up an EFS instance, you can add add files without worrying about provisioning or disturbing your application's performance
- Additional accessibility – EFS can be mounted from different EC2 instances, but it can also cross the region boundary via the use of VPC peering
- **When to use EFS?** EFS may be used whenever you need a shared file storage option for multiple EC2 instances with automatic, high-performance scaling. This makes it a great candidate for file storage for content management systems, storing code and media files.

Object storage Vs File storage Vs block storage

- **Block storage :** In Block storage, data is divided into smaller blocks and stored across multiple physical drives. Each block has a unique address, making block storage ideal for high-performance applications like databases.
- **Object storage's** rich metadata and scalability make it more suitable for environments that handle **vast amounts of unstructured data**.
- **File Storage** may be used whenever you need a shared file storage option for multiple instances with automatic, high-performance scaling.

Amazon S3 Vs Amazon EBS Vs Amazon EFS

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	S3(Object)	EBS(Block)	EFS(File System)
Type of storage	Object storage. You can store virtually any kind of data in any format.	Persistent block level storage for EC2 instances.	File storage for EC2 instances.
Features	Accessible to **anyone** or any service with the right permissions	Delivers performance for workloads that require the lowest-latency access **from a single EC2 instance**	Has a file system interface, and **concurrently-accessible** storage for multiple EC2 instances
Max Storage/ Max file size	S3 can store large amounts as compared to EBS. With S3, the standard limit is of 100 buckets and each has got an unlimited data capacity. Max object size 5TB.	EBS has a standard limit of 20 volumes and each volume can hold data up to 1TB. In EBS there occurs an upper limit on the data storage. Max file size = volume size. Default size of a block is 4KiB.	Unlimited system size. 47.9 TiB for a single file

Amazon S3 Vs Amazon EBS Vs Amazon EFS

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	S3	EBS	EFS
Performance (Latency)	Low, for mixed request types, and integration with CloudFront	Lowest, consistent; SSD-backed storages include the highest performance Provisioned OPS SSD and General Purpose SSD that balance price and performance.	Low, consistent; use Max I/O mode for higher performance
Durability	Stored redundantly across multiple AZs; has 99.999999999% durability	Stored redundantly in a single AZ	Stored redundantly across multiple AZs
Scalability	Highly scalable	Manually increase/decrease your storage size. Attach and detach additional volumes to and from your EC2 instance to scale.	EFS file systems are elastic, and automatically grow and shrink as you add and remove files.

Amazon S3 Vs Amazon EBS Vs Amazon EFS

	S3	EBS	EFS
Data Accessing	One to millions of connections over the web; S3 provides a REST API web services interface	Single EC2 instance in a single AZ. Amazon EBS Multi-Attach volume to up to 16 Nitro-based instances that are in the same AZ.	One to thousands of EC2 instances or on-premises servers, from multiple AZs, regions.
Pricing	S3 allows you to follow the utility-based model and prices as per your usage . Free Tier – 5 GB First 50 TB/month – $0.023/GB	In EBS, you need to pay for the provisioned capacity. Free Tier – 30 GB, 1 GB snapshot storage General Purpose – $0.045/GB(1 month)	Varies from .33$ TO .01$ GB-Month Depending on Standard, Infrequent access or Archive types.
Use Cases	Web serving and content management, media and entertainment, backups, big data analytics, data lake	Boot volumes, transactional and NoSQL databases, data warehousing & ETL	Content management, enterprise apps, media and entertainment, home directories, database backups, developer tools, container storage, big data analytics

Object and Block Storage Use cases

Object Storage use cases are:

1. Data lake and big data analytics:
2. Backup and restoration:
3. Reliable disaster recovery
4. Other use cases include entertainment, media, content management purposes, etc.
5. Long term storage, archiving, Cold storage

Block storage use cases are:

1. Software testing and development
2. Business continuity
3. Enterprise-wide applications
4. Transactional and NoSQL databases

File System Use Cases: Content management, enterprise apps, media and entertainment, home directories, database backups, developer tools, container storage, big data analytics

Instance store - temporary block storage for VMs

An ***instance store*** provides temporary block-level storage for your VM instance. This storage is provided by disks that are physically attached to the host computer. Instance store is ideal for temporary storage of information that changes frequently, such as buffers, caches, scratch data, and other temporary content. It can also be used to store temporary data that you replicate across a fleet of instances, such as a load-balanced pool of web servers. User applications can utilize Instance Store for temporary storage needs. It's ideal for workloads that require low-latency access to data, such as caching, session management, or temporary storage for processing large datasets/

The life of an instance store is the same as the life of the instance it's attached to. This means that data in an instance store is only available while the instance is active.

- An instance store is a type of temporary block-level storage for a VM instance.
- It's a local disk drive for the virtual machine.
- Data in the instance store persists as long as the instance is active, even if it reboots.
- When the instance hibernates or terminates, all data in the instance store is lost.